

Statement of Purpose
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Master's Program in Health Services Management and Hospitals
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The first problem that shaped my academic direction was not a medical problem. It was a decision problem. At 2ACOM, while building a five-year profitability model for outdoor advertising assets in Casablanca, I saw how easily an investment could appear attractive when judged through one visible figure, then become more complex once traffic exposure, location quality, CAPEX, OPEX, break-even timing, and regulatory constraints were placed in the same model. At MAGHREBAIL, portfolio analysis taught me the same lesson from another angle: risk is rarely visible in the headline amount; it emerges through commitments, delays, exposure, and weak monitoring. At SAN MAROC, working with financial-performance dashboards and recovery-priority logic confirmed the pattern. Better decisions require more than recording what is immediately measurable. They require a framework that makes hidden cost, operational risk, and long-term exposure visible before the decision is made.

This is the intellectual path that leads me to the Master's Program in Health Services Management and Hospitals at King Abdulaziz University. My interest is not clinical practice. It is the administrative and financial architecture that determines whether hospitals can sustain service quality under budget, procurement, and lifecycle-cost constraints. Public hospitals do not only need skilled clinicians; they also need managers capable of evaluating costly assets, structuring procurement choices, monitoring supplier risk, and translating financial sustainability into operational decisions.

My academic background prepared me for this direction. I completed a Professional Bachelor's Degree in Management, majoring in Financial and Accounting Management, at the International University of Casablanca. My training in accounting, corporate finance, management control, financial analysis, statistics, economics, business law, and organizational management gave me the technical base to understand organizations through cost structures, performance indicators, and decision constraints. I am currently pursuing graduate work in Procurement and Supply Chain Management at INSEEC Business School in Lyon, where my focus has expanded toward procurement performance, lifecycle-cost visibility, supplier evaluation, forecasting, KPI modelling, and data-supported managerial decision-making.

My bachelor final project became the first major test of this analytical orientation. I studied the profitability of outdoor advertising infrastructure in Casablanca by linking road traffic, location value, regulation, CAPEX, OPEX, TCO, ROI, break-even analysis, and profitability scoring. The project required more than a standard finance calculation. I had to connect spatial exposure, simulated traffic flows, pricing assumptions, asset costs, and strategic location decisions. The main lesson was methodological: a financially defensible recommendation must combine cost, utilization, risk, and context. A low-cost asset is not necessarily the best asset. A high-cost asset is not necessarily irrational. The correct question is whether the full lifecycle value justifies the commitment.

I later extended this reasoning through an AI-MCDM research manuscript on smart-city infrastructure investment under carbon, energy, and geoeconomic risk. In that work, I developed an AI-enabled multi-criteria decision-analysis scorecard combining XGBoost forecasting, carbon-adjusted valuation logic, real-options reasoning, Fuzzy-AHP/MEREC weighting, and AROMAN ranking. This project strengthened my interest in transparent decision-support systems. I learned that advanced models are only useful when their assumptions are visible, their criteria are traceable, and their results are tested under alternative scenarios. This principle now guides the research I intend to pursue at KAU.

My proposed research topic is: **"A Discounted Total Cost of Ownership and BWM-TOPSIS Framework for Value-Based Procurement of High-Cost Medical Devices in Saudi Public Hospitals."** The problem is concrete. A hospital may select a medical device because its acquisition price appears favorable at tender award, but the real cost may appear later through maintenance contracts, proprietary consumables, spare parts, staff training, software upgrades, downtime, facility adaptation, disposal, and residual-value uncertainty. A procurement decision that is rational on the first day can become inefficient over the device lifecycle if hidden costs and operational risks are not evaluated systematically.

My proposed framework is deliberately non-clinical. I do not intend to assess patient outcomes, biomedical performance, clinical effectiveness, or confidential supplier bids. Instead, I aim to build a hospital-management decision-support model that combines discounted Total Cost of Ownership, Best-Worst Method weighting, TOPSIS ranking, and sensitivity analysis. The model would evaluate high-cost medical-device alternatives through lifecycle affordability, service continuity, supplier reliability, digital readiness, governance alignment, and sustainability. Expert judgment from healthcare finance, procurement, hospital administration, supply-chain, biomedical-engineering-management, and health-economics specialists would validate the criteria and generate weights. Synthetic procurement scenarios would then allow the model to rank alternatives and test whether the ranking remains stable when assumptions change.

This topic fits King Abdulaziz University because it sits exactly between health administration, economics, finance, and public-sector management. The Department of Health Services and Hospitals Management offers the academic environment I need to convert my financial and procurement background into specialized competence in hospital administration. KAU's location in Saudi Arabia is also essential to the project. Saudi Vision 2030 and the Health Sector Transformation Program create a policy context where hospital performance, quality improvement, digital transformation, and financial sustainability are not abstract goals; they are national priorities. In that environment, procurement is not a back-office function. It is a strategic mechanism through which public hospitals protect budgets, maintain service continuity, and improve accountability.

NUPCO's role in Saudi healthcare procurement further reinforces the relevance of my proposed research. Centralized procurement can improve efficiency and standardization, but procurement committees still need transparent tools to compare device alternatives beyond acquisition price. My study would not replace institutional procurement systems or claim access to confidential national data. Its contribution would be academic and managerial: a reproducible framework that helps decision-makers structure trade-offs, document assumptions, and evaluate procurement alternatives through both cost and non-price criteria.

My professional experiences have prepared me for this work without giving me a false clinical identity. At MAGHREBAIL, I analyzed client portfolios, commitments, leasing-finance indicators, and credit-risk exposure. That experience trained me to read financial risk as a pattern, not as an isolated number. At 2ACOM, I converted traffic exposure and asset-cost evidence into investment-location recommendations. That work taught me how to move from raw indicators to a ranked decision. At SAN MAROC, I worked on financial-performance analysis, portfolio segmentation, recovery indicators, and dashboard automation. That experience strengthened my ability to convert complex financial information into decision-support outputs for managers.

After completing the Master's degree, my objective is to work in healthcare procurement, hospital performance management, or health-sector consulting, with a focus on lifecycle costing, value-based purchasing, procurement governance, and data-supported resource allocation. In the longer term, I aim to contribute to healthcare systems in Saudi Arabia, Morocco, and the wider MENA region by developing decision-support tools that help public institutions spend better, not merely spend less.

I am applying to KAU because my academic trajectory now requires a program where hospital administration is treated as a rigorous management discipline. I bring a finance and procurement profile, experience in risk analysis and KPI reporting, and a research agenda grounded in lifecycle cost and multi-criteria decision-making. I understand the limits of my non-clinical background, but I also understand its value. Hospitals need administrators who can make complex resource decisions visible, auditable, and financially sustainable. That is the contribution I intend to develop at King Abdulaziz University.